

Machine Learning Career Roadmap

A practical 12–18 month roadmap for becoming an ML Engineer

Target: Software Engineering + Machine Learning + MLOps + GenAI

Phase	Focus	Approx. Time
1	Python + Mathematics	1–2 months
2	Data Analysis	1 month
3	Classical Machine Learning	2–3 months
4	ML Projects	2 months
5	Deep Learning with PyTorch	2–3 months
6	Choose a Specialization	1–2 months
7	GenAI / LLMs	1–2 months
8	MLOps & Deployment	Ongoing

Phase 1 — Python + Math

- Python: functions, OOP, collections, files, exceptions, virtual environments.
- Libraries: NumPy, Pandas, Matplotlib.
- Math: vectors, matrices, dot product, probability, statistics, derivatives and gradients.
- Goal: understand the mathematics behind ML without trying to become a mathematician.

Phase 2 — Data Analysis

- Learn data cleaning, missing values, duplicates, outliers and categorical variables.
- Practice exploratory data analysis (EDA), distributions, correlations and visualization.
- Goal: take a messy dataset and extract useful insights from it.

Phase 3 — Classical Machine Learning

- Supervised learning: Linear Regression, Logistic Regression, KNN, Decision Trees, Random Forest, Gradient Boosting, XGBoost and SVM.
- Unsupervised learning: K-Means, Hierarchical Clustering and PCA.
- Core concepts: train/test split, validation, cross-validation, overfitting, underfitting, bias/variance and feature engineering.
- Evaluation: precision, recall, F1-score and ROC-AUC.
- Main library: scikit-learn.

Phase 4 — ML Projects

- Student Performance Predictor — predict marks from study hours, attendance and previous performance.
- Customer Churn Prediction — predict whether a customer is likely to leave.
- Fraud Detection — classify transactions as fraudulent or legitimate and learn about imbalanced datasets.
- Focus on clean README files, reproducible code, metrics and explanations.

Phase 5 — Deep Learning

- Neural networks, backpropagation, loss functions and optimization.
- CNNs, RNN/LSTM concepts and Transformers.
- Use PyTorch: tensors, datasets/dataloaders, nn.Module, training loops, validation and GPU training.
- Learn transfer learning and model saving/loading.

Phase 6 — Choose a Specialization

- **Computer Vision:** CNNs, transfer learning, YOLO, object detection, segmentation and vision transformers.
- **NLP/LLM:** NLP fundamentals, embeddings, Transformers, BERT, LLMs, RAG and agents.
- **Recommendation Systems:** collaborative filtering, content-based methods, embeddings and ranking.
- Choose one primary specialization rather than trying to master everything at once.

Phase 7 — GenAI / LLMs

- Learn LLM APIs, embeddings, vector databases, RAG, prompt engineering, fine-tuning, LoRA/QLoRA, evaluation and agents.
- Typical RAG pipeline: document → text extraction → chunks → embeddings → vector database → retrieval → LLM → answer.
- Do not skip ML and deep-learning fundamentals just because GenAI is popular.

Phase 8 — MLOps & Deployment

- Learn Git/GitHub, Linux, Docker, REST APIs, FastAPI, CI/CD, MLflow, cloud deployment and model monitoring.
- Typical production flow: train model → save model → expose API → Dockerize → deploy → monitor.
- This is especially valuable for developers who already enjoy backend and DevOps.

Recommended Daily Routine

- **1 hour — DSA:** Continue LeetCode and problem-solving practice.
- **1 hour — ML:** Study concepts, mathematics and implementation.
- **1 hour — Project:** Build, test and document something real.

Suggested 2026–2027 Path

Period	What to Learn
2026	Python → NumPy/Pandas → Statistics → Classical ML → SQL
Late 2026	scikit-learn → ML projects → PyTorch → Deep Learning
2027	Specialization → LLM/RAG → FastAPI → Docker → Cloud → MLOps
Afterward	Apply for ML/AI internships and entry-level ML/AI roles

Portfolio Structure

A strong GitHub portfolio can be organized around foundations, projects, deep learning, GenAI and MLOps:

ml-foundations/ • ml-projects/ • deep-learning/ • genai/ • mlops/

Aim for 5–7 strong projects rather than many copied projects. Each project should explain the problem, dataset, approach, metrics, limitations and how to run it.

Career Roles

- ML Engineer / Junior ML Engineer
- AI Engineer / Applied ML Engineer
- Data Scientist
- Computer Vision Engineer
- NLP Engineer
- GenAI Engineer

Important Principle

Do not become only an API user. Build the foundation: Python → Math/Statistics → Data → Classical ML → Deep Learning → LLMs → Deployment → MLOps.

For a CSE student, the strongest combination is software engineering + ML + deployment. Existing backend, Linux, Git, cloud and DevOps skills can become a major advantage when building production AI systems.